Byung-cheol Kang

Bioinformatics Research & Development · Genomics · Bacteriophages

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PROFILE

A bioinformatics researcher and developer connecting biological questions with analysis design and interpretation. My work extends from microbes to humans, integrating diverse analytical approaches and building research pipelines and databases. AI-powered coding accelerates my research and amplifies my ability to automate workflows and develop analytical tools.

EXPERIENCE

MicrobiotiX · Senior Researcher · Bioinformatics — 2026.07 - PRESENT

- Analyze bacterial and phage genomes to prepare candidate-assessment reports and evidence for follow-up research.
- Develop tools to select and compare phage cocktail candidates from measured host-range data.
- Build a capsule-target reference database from 7,759 collected and clustered tail/RBP sequences; develop and evaluate prediction libraries and ML models.
- Compare RBP candidates for target capsule types and generate candidate sequences supported by sequence and protein-structure evidence.

Wittgenbio · Computational Biology Research — 2026.07 - 2026.10

- Built a multi-study CRC scRNA-seq atlas and performed cell-type annotation.
- Cross-checked tumor/normal metadata, author annotations and marker evidence, with versioned label and source tracking.
- Annotated externally supplied normal-tissue atlas data and compared normal-tissue, tumor and patient/specimen expression to assess therapeutic-target candidates.

Sanigen · Bioinformatics Team Lead · Senior Researcher — 2020.03 - 2025.11

- Led NGS analysis services and team operations, including analysis design, hands-on work, result review, SOPs, turnaround management and quality control.
- Performed genomics, microbiome and transcriptome analyses; automated workflows and reports and resolved analysis errors.
- Developed comparative-genomics and k-mer marker-discovery methods and contributed to collaborative NGS pathogen-panel research.
- Contributed to the technology-special listing task force, preparing and revising technical-evaluation and preliminary-listing application materials.
- Delivered custom analyses and method evaluations for external collaborators, client technical support and training materials.
- Maintained HPC/Linux analysis environments, NAS storage and data-transfer infrastructure.

SELECTED PROJECTS

PHAGE DATABASE AND DESIGN

Klebsiella Phage Reference Database & In Silico Design

Built a tail/RBP and KL-type reference database to connect protein sequences with capsule-target evidence. Integrated anchor compatibility, reference-label consistency, and protein-structure evidence to prioritize in silico tail-RBP design candidates for experimental follow-up.

REPRODUCIBLE ANALYSIS PLATFORM

Sanigen Integrated Microbial Genomics Platform

Designed and maintained a shared Snakemake platform for 16S, ITS, WGS, and Illumina-ONT hybrid assembly. Standardized inputs, execution environments, QC checkpoints, logs, and report packaging to support reproducible analysis and review.

SINGLE-CELL AND TISSUE EXPRESSION

CRC Atlas Annotation & Normal-Tissue Expression Analysis

Integrated CRC datasets across studies, annotated cell types, and reviewed annotation consistency and tumor/normal labels. Separately annotated externally supplied normal-tissue atlas data and compared tissue-specific expression to evaluate candidate therapeutic targets, accounting for sequencing-depth differences and annotation coverage.

MARKER DISCOVERY

Comparative Genomics-Based Marker Discovery

Compared complete microbial genomes to identify candidate marker regions and assess their specificity, conservation, and suitability for downstream assays. Prioritized an approximately 17 kb region for experimental follow-up.

TECHNICAL SKILLS

Genomics	Bacterial WGS, phage genomics, assembly, annotation, comparative genomics, KL typing
Phage and protein	Tail/RBP analysis, depolymerase curation, structure comparison (Foldseek)
Single-cell analysis	CRC atlas construction, cell-type annotation, normal-tissue expression analysis
Workflow engineering	Reproducible analysis workflows and execution environments (Snakemake)
Programming	Python, R, Bash, Git
Phage annotation	Pharokka

EDUCATION

M.S., Medical Science · Kyungpook National University School of Medicine · 2016.02 - 2018.08
B.S., Biotechnology; Minor in Psychology · Kyungpook National University · 2008.03 - 2016.02

SELECTED PUBLICATIONS & IP

- Park et al. Novel next generation sequencing panel method for multiple detection and identification of foodborne pathogens in agricultural wastewater. **Frontiers in Microbiology** 14 (2023).
- Park et al. Development and evaluation of a next-generation sequencing panel for pathogens in fermented foods. **Journal of Microbiology and Biotechnology** 33(1) (2023).
- Kang et al. Nothofagin suppresses mast cell-mediated allergic inflammation. **Chemico-Biological Interactions** 298 (2019).
- Joint patent contribution: NGS-based primer set and detection method for foodborne bacteria (2022).

LANGUAGE

Korean - Native English - IELTS 6.5 (October 2023)